

AI-Powered Retail Business Intelligence Report

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Analysis Mode: anomaly

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1. Anomaly Detection Results

Total records analyzed: 336

Total anomalies detected: 17

Anomaly rate: 5.06%

Risk Distribution:

- Low Risk: 17

Anomaly Type Distribution:

- Inconsistent Demand Pattern: 17

Highest Risk Dates:

- Not available

Business Interpretation:

The LSTM Autoencoder learns normal temporal retail behavior. Anomalies are detected using reconstruction error and business rules.

Model Execution Evidence:

- Loaded model: outputs/best_anomaly_lstm_autoencoder.pt
- Execution mode: inference using saved trained model
- Input dataset was automatically preprocessed
- Anomalies were detected using reconstruction error analysis
- The LSTM Autoencoder evaluated temporal retail behavior patterns

2. Business Recommendations

[Medium] Anomaly Monitoring

Recommendation: Investigate abnormal sales and promotion behavior before making final stock decisions.

Reason: 17 anomalies were detected with risk distribution: {'Low Risk': 17}.

[Medium] Operational Risk

Recommendation: Review the main anomaly categories to identify operational inconsistencies.

Reason: Detected anomaly types: {'Inconsistent Demand Pattern': 17}.

[High] Business Control

Recommendation: Validate suspicious periods before taking inventory or promotion decisions.

Reason: Human review is required because abnormal behavior may indicate stock or promotion issues.

3. CrewAI Business Insight

SUMMARY:

- The anomaly detection model identified 17 anomalies out of 336 analyzed records.

KEY FINDINGS:

- The anomaly rate is 5.06%.
- The main anomaly type is Inconsistent Demand Pattern.
- Risk distribution: {'Low Risk': 17}.

MAIN RISK:

- Abnormal sales or promotion behavior may lead to wrong inventory, promotion, or operational decisions.

RECOMMENDED ACTIONS:

- Investigate abnormal sales and promotion behavior before making final stock decisions.
- Review the main anomaly categories to identify operational inconsistencies.
- Validate suspicious periods before taking inventory or promotion decisions.

4. CrewAI Executive Summary

EXECUTIVE SUMMARY:

- The anomaly analysis detected abnormal retail behavior that requires business review.

MODEL RESULT:

- 17 anomalies were detected out of 336 records.
- The anomaly rate is 5.06%.
- The main anomaly type is Inconsistent Demand Pattern.

BUSINESS RECOMMENDATIONS:

- Investigate abnormal sales and promotion behavior before making final stock decisions.
- Review the main anomaly categories to identify operational inconsistencies.
- Validate suspicious periods before taking inventory or promotion decisions.

HUMAN VALIDATION:

- Human validation status: Validated.

FINAL DECISION:

- Review detected anomalies before making inventory or promotion decisions.

5. Human Validation

Final status: Validated

Total recommendations: 3

Approved recommendations: 3

Rejected recommendations: 0

Human validation required: True

6. Final Conclusion

The AI system analyzed retail data, generated business recommendations, and validated decisions through a human-in-the-loop checkpoint.